

# Gabriel Talbert Bunt

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Computer Science graduate of Stevens Institute of Technology (BSc, 2025) specializing in embodied AI and autonomous systems. Developed assistive technology using egocentric vision for visually impaired users' form-filling tasks. Currently optimizing drone path planning through multi-modal data integration. Self-studied in neural radiance fields, stereo matching, and core computer vision concepts. Seeking doctoral research to advance autonomous learning and develop more robust, adaptable AI systems.

## Education

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### Stevens Institute of Technology

Expected May 2025

*Bachelor of Science in Computer Science*

GPA 3.97/4.0, John V. Ernest '50 Scholarship, Edwin A. Stevens Scholarship, Stevens Grant.

President of Stevens Linux Users Group.

Relevant Coursework:

- Machine Learning Fundamentals: Explored the mathematics behind machine learning.
- Deep Learning: Implemented logistic regression, CNN, LSTM, and supervised autoencoder.
- Computer Vision: Implemented filtering, corner detection, template matching, Hough transform, ...

## Research

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### Optimizing Swarm Trajectories

September 2024 – Present

*supervised by Ying Wang*

- Working in a team of 6 to predict optimal drone trajectories of a swarm.
- Integrating multi-modal sensor data to localize aerial vehicles relative to each other and the world.
- Collecting large volumes of data from a variety of sensors.

### Enabling Blind Persons to Write Independently

June 2024 – Present

*supervised by Jonggi Hong*

- Developed an innovative handwriting-assistance tool for blind persons using computer vision and AI techniques.
- Implemented localization using MediaPipe and YOLO to track non-dominant index fingertip and pen tip positions.
- Designed and conducted a Wizard of Oz study to test the feasibility and effectiveness of the approach.
- Achieved fingertip tracking accuracy within 1cm using MediaPipe, suitable for form navigation.
- Identified challenges in pen tip tracking and proposed future improvements through reproducing DodecaPen.
- Presented findings through a poster at the Stevens Institute of Artificial Intelligence AIRS Fellowship Poster Session.

## Teaching Experience

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### Tutor

April 2024 – Present

*Independent*

- Provide personalized one-on-one instruction in physics and calculus, emphasizing foundational conceptual understanding.
- Successfully helped two struggling students improve their grades from C/D range to B+ or higher.
- Incorporate real-world applications and interdisciplinary connections to enhance engagement and retention of key concepts.

## **Tutor**

January 2023 – Present

*Stevens Institute of Technology*

- Instruct 50+ students through one-on-one and small group sessions across advanced CS topics including data structures, algorithms, machine learning, and systems programming, as well as foundational math courses.
- Emphasize development of intuitive understanding and problem-solving skills, enabling students to apply concepts across various scenarios rather than memorizing specific details.

## **Course Assistant, Intro to Computer Science**

September 2023 - December 2023

*Stevens Institute of Technology*

- Directed hands-on programming labs for 30+ students weekly, focusing on practical application of theoretical concepts.
- Provided targeted one-on-one guidance during office hours, helping students overcome coding challenges and deepen their understanding of CS fundamentals.

## **Tutor**

February 2022 – August 2022

*Skagit Valley Community College*

- Tutored 20+ students in calculus, English, Spanish, and introductory computer science.
- Adapted teaching methods to individual learning styles, ensuring comprehensive understanding of complex concepts.

## **Funding**

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- SIAI AIRS Fellowship

## **Projects**

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### **Stereo Matching Algorithms**

April 2024 – June 2024

*[github.com/GabrielDTB/stereo-matching](https://github.com/GabrielDTB/stereo-matching)*

- Implemented stereo pixel matching algorithms using Sum of Absolute Differences (SAD) along epipolar lines, with support for Census and Rank transforms.
- Developed a disparity calculation pipeline to compute pixel-wise depth information from stereo image pairs, enabling 3D scene reconstruction.
- Optimized performance through parallel processing using Rayon, achieving significant speedup on multi-core CPUs.
- Formulated evaluation metrics such as error rate, error mapping, and validity mapping.
- Created a command-line interface using Clap for easy parameter adjustment and file I/O handling.

## **References**

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- Philippos Mordohai
- Jonggi Hong
- Sandeep Bhatt